What is the Internet?

↳ Nuts & bolts View:

1. Host devices/End Systems that run network apps

2. Packet Switches that forward chunks of data (packets)

   There are two types of packet switches routers & switches.
   ↳ Under both MAC & IP address   ↳ understand only MAC address

3. Communication links that interconnect routers, switches, hosts & end-systems.

   These include: fibre, copper, radio, satellite

These Host devices, packet switches & links are assembled into networks, each of which are owned & operated by entities.

It is the existence of these multiple networks that gives rise to the saying that internet is network of networks.

Sending & recieving information among routers, switches, hosts & end devices is controlled by protocols.

Internet Engineering Task Force defines standards that protocols must adhere to.

↳ A Services View:

Internet provides a platform that applications can send & recieve information.

The applications are said to be distributed applications, since they involve multiple end systems that exchange data with each other.

Applications run on end systems & not packet switches

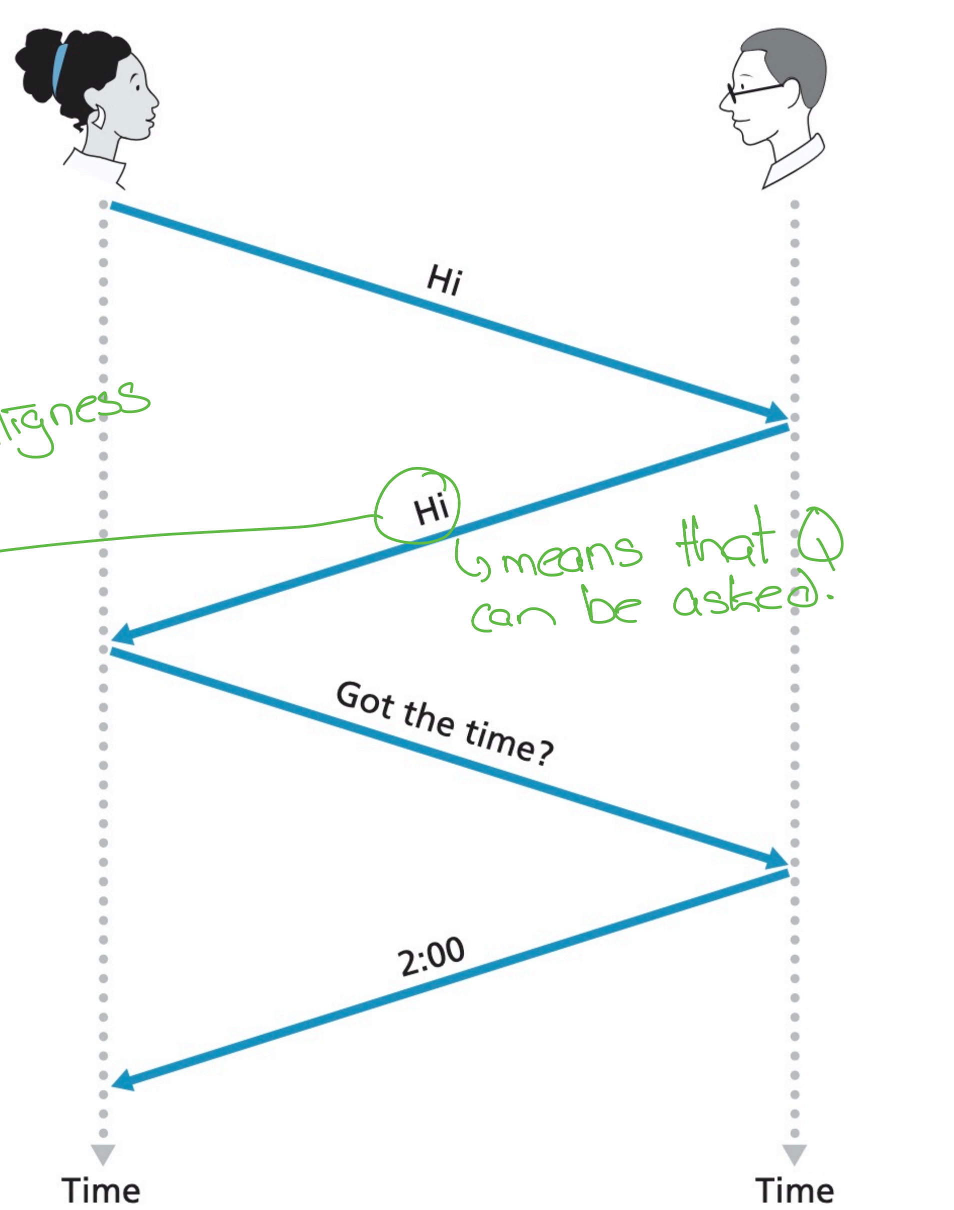
Suppose you develop an Application. Now, how does one program running on one end system instruct the internet to deliver data to another program running on another end-system?

Internet provides a Socket Interface.

It is essentially a set of rules that sending program must follow to so that internet can deliver data to destination program.

What is a Protocol?

Human Analogy: Consider when you ask someone time of the day.



An alternate can be "Don't bother me" might indicate unwillingness to communicate

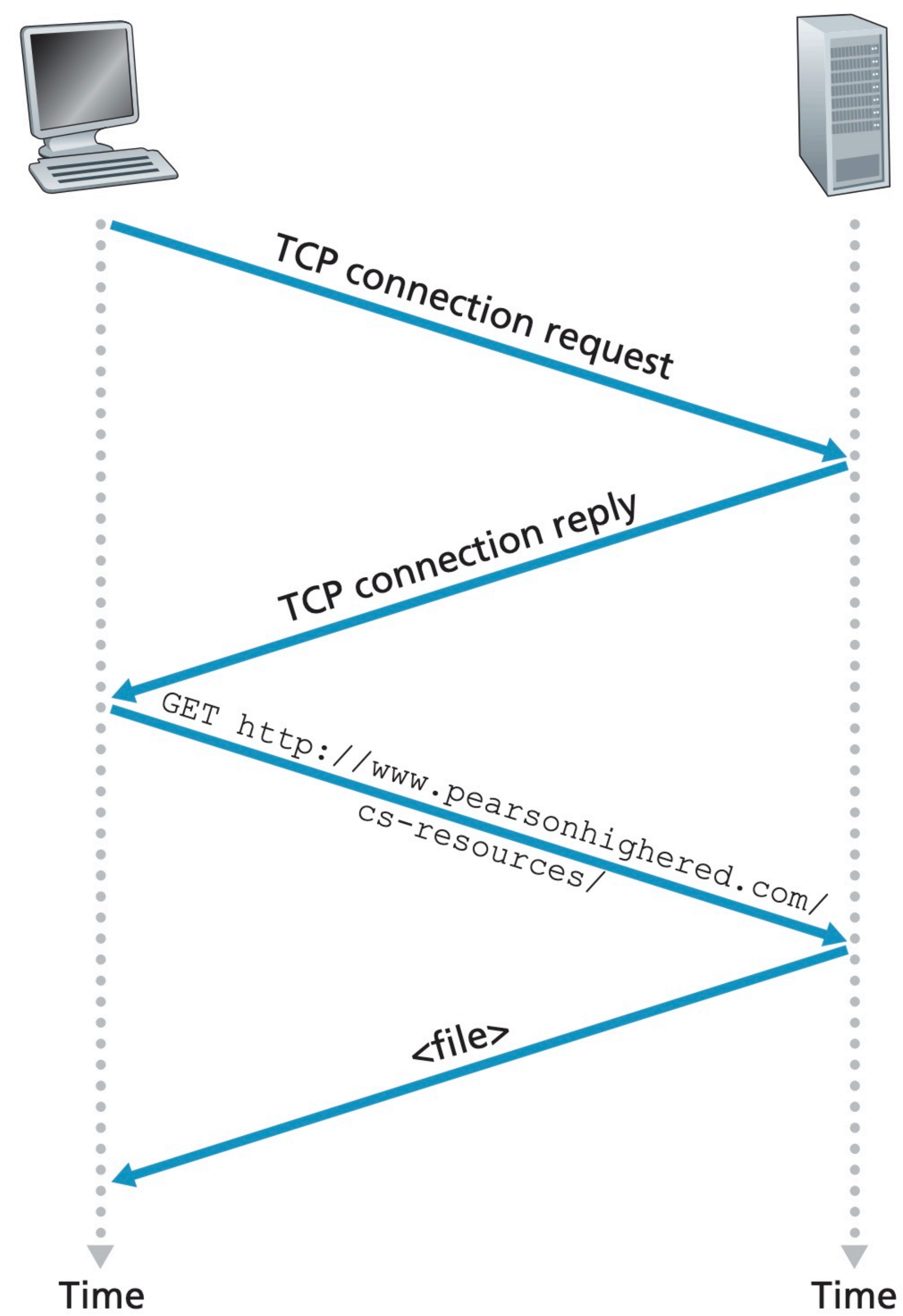
↳ means that Q can be asked.

↳ In our human protocol, there are specific messages we send and specific actions we take in response to the recieved reply messages or other events.

Network Protocols

In network protocols, entities are hardware or software components of some device.

For example,



Thus, a protocol defines the format & the order of messages exchanged b/w two or more communicating entities, as well as the actions taken on the transmission and/or reciept of a message or other event.

The Network Edge

↳ Network Edge comprises of end systems / hosts

because they sit at edge of internet

because they appli- cation programs.

Hosts can be divided into:

1. Clients:
desktops, laptops, smartphones

2. Servers
Powerful machines that store & distribute web pages.

Access Networks (2)

The network that physically connects an end system to the first router/edge router on path from the source to destination.

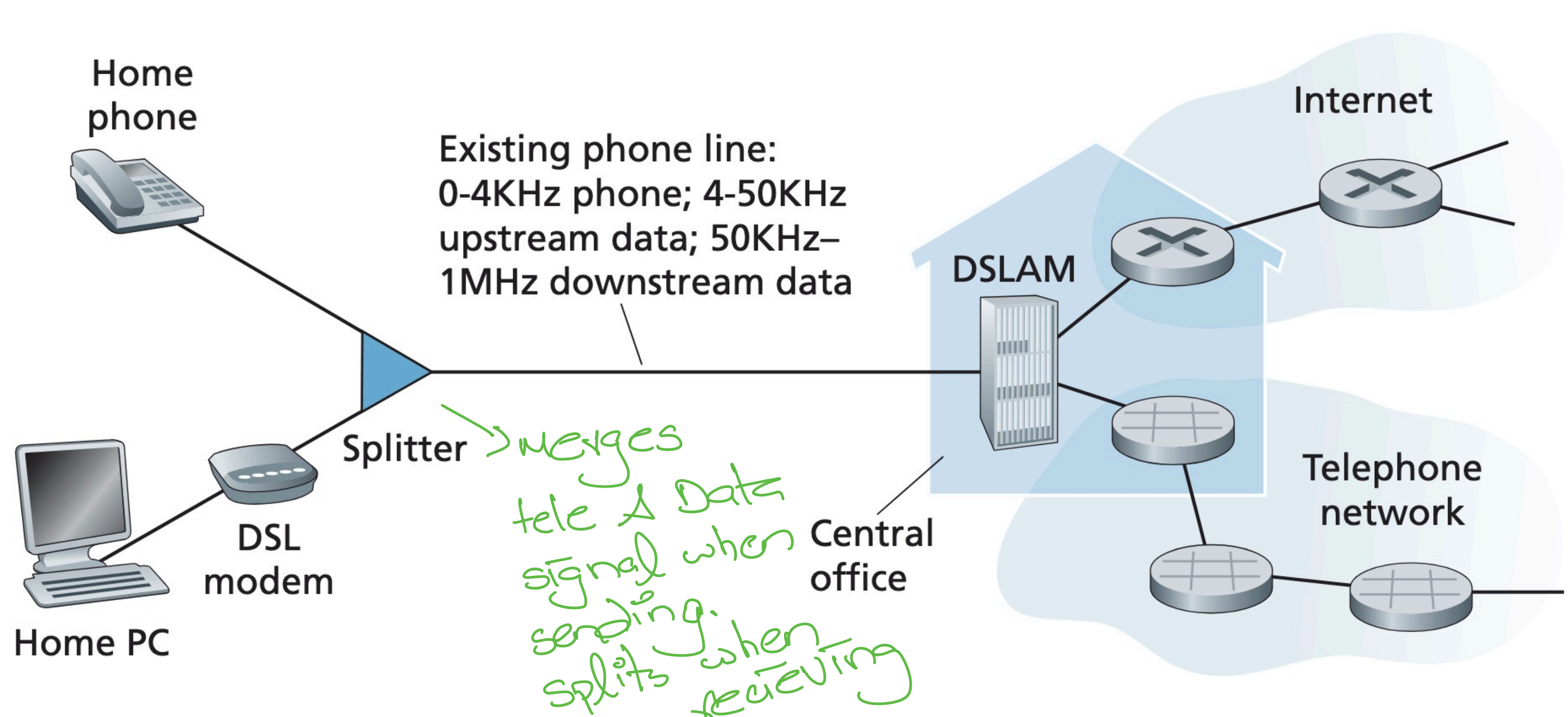
OR the network that connects host/end system to global internet.

There are several types of access networks: home, institutional, mobile access networks.

Home Access

Digital Subscriber line

- A residence obtains DSL internet access from the same local telephone company (telco) that provides its wired phone access.



As shown in figure, each customer's DSL modem uses the existing line exchange data with a digital subscriber line access multiplexer (DSLAM).

The home's DSL modem takes digital data & translates it to high frequency tones for transmission over telephone wires to the CO; the analog signals from many such houses are translated back into digital format at DSLAM.

The telephone line carries both data & telephone signals simultaneously, which are encoded at different frequencies:

↳ If high freq, then low range

- A high-speed downstream channel, in the 50 kHz to 1 MHz band
- A medium-speed upstream channel, in the 4 kHz to 50 kHz band
- An ordinary two-way telephone channel, in the 0 to 4 kHz band

This makes the single DSL appear as three separate links.

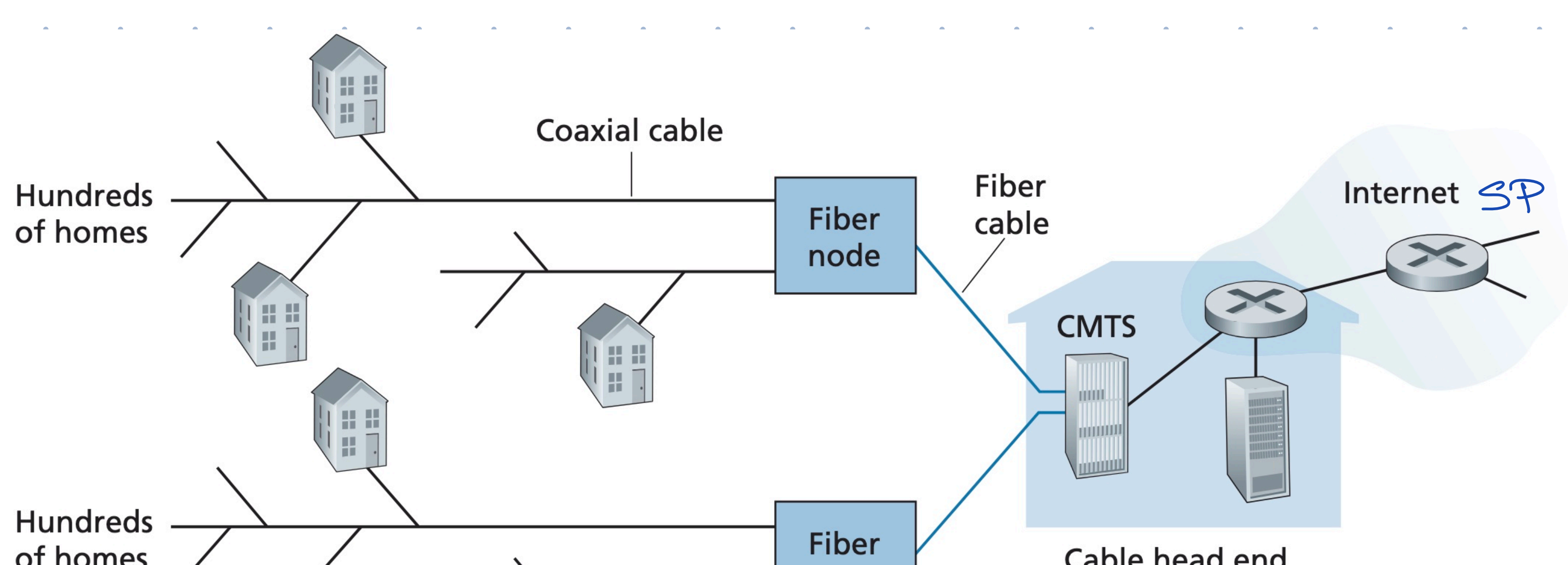
On customer side, a splitter separates telephone & Data signals arriving & forwards to the DSL modem. This is called frequency division multiplexing.
On the other hand, there is time division multiplexing which acts like RR
This has Good bandwidth but bad latency.

On telco side, DSLAM separates data & phone signals & sends the data into internet.

DSL is designed for short distances b/w home & CO.
Downstream & Upstream happen on different freqs allowing simultaneous.

Cable-based Internet Access

This makes use of the cable tv company's existing cable tv infrastructure



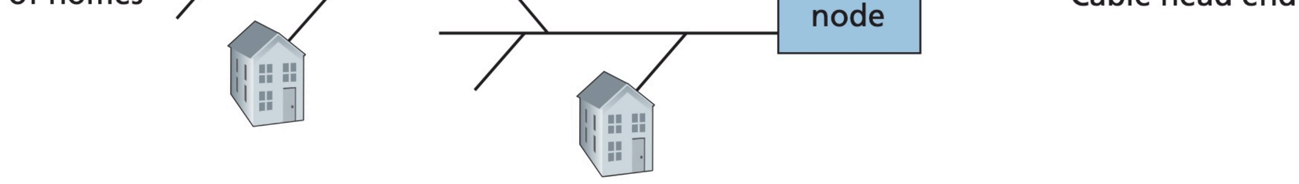


Figure 1.6 ♦ A hybrid fiber-coaxial access network

As shown in the figure, a fibre cable is used to connect cable head end to neighbourhood-level junctions.

Each neighbour junction uses coaxial cables to connect 500 - 5000 homes.

As both fibre & coaxial cables are used, it is often referred Hybrid fibre coax.

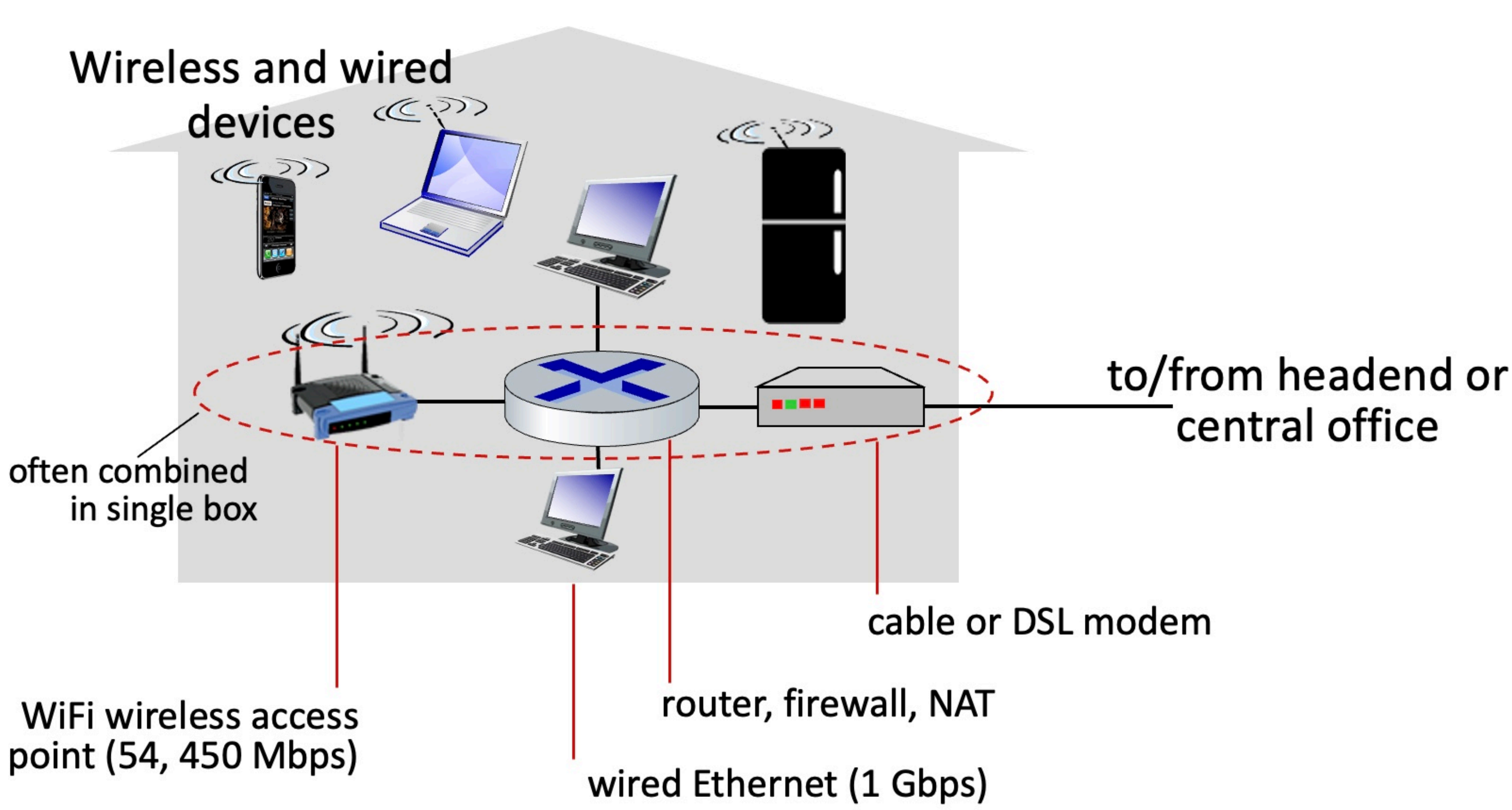
Cable internet Access requires special modems, called cable modems, which to the connects home PC through Ethernet.

Cable Modem termination System (CMTS) serves similar function as DSLAM.

Access networks:

Home Networks

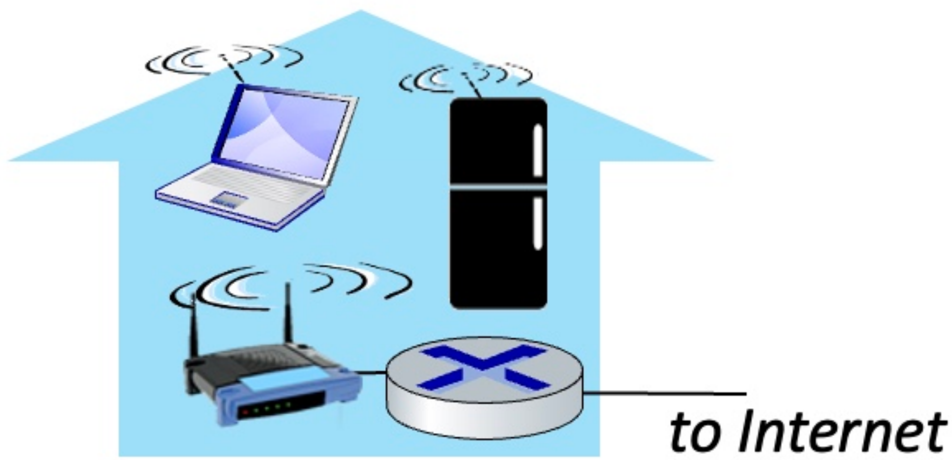
A typical home network looks like:



Wireless Access Networks

There are two classes of wireless networks

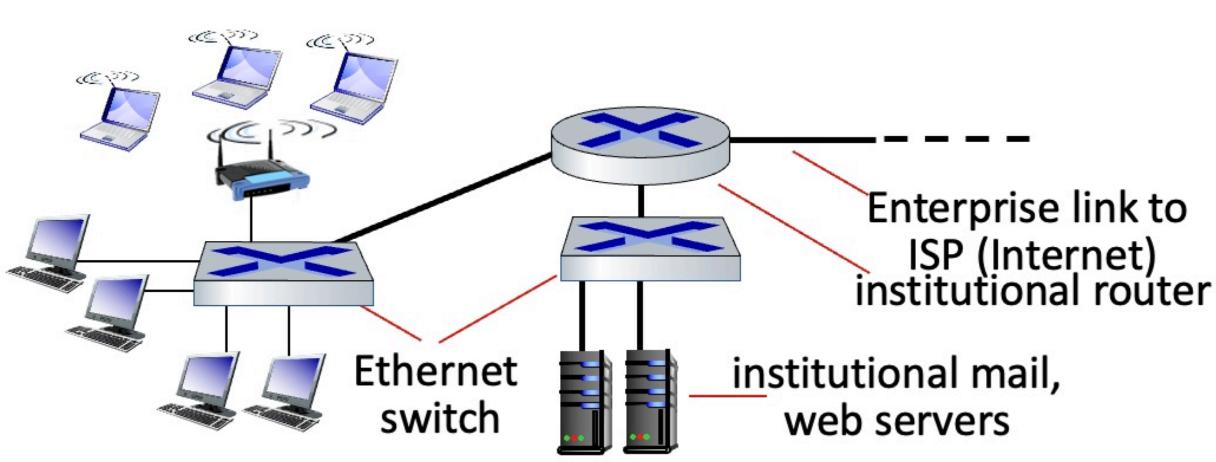
- Wireless local Area Networks (Wifi)
  - ↳ Within or around building
  - ↳ 802.11b/g/n (Wifi) : 11, 54, 450 Mbps transmission rate



- Wide-area cellular access networks (3G, 4G, 5G)
  - ↳ Provided by mobile, cellular network operator

For both classes, there's an entity / a base station / an access point to which end devices transmitting data to & recieving data from

Enterprise Networks



↳ Home Networks on Steriods.

↳ Has mix of wired, wireless.

↳ Typically has large no. of switches & routers.

Another type of Enterprise Networks are data center networks:

↳ Connects massive no. of servers to each other & to the internet.

Packets

↳ Host has some data, it wants to send, say a large file.

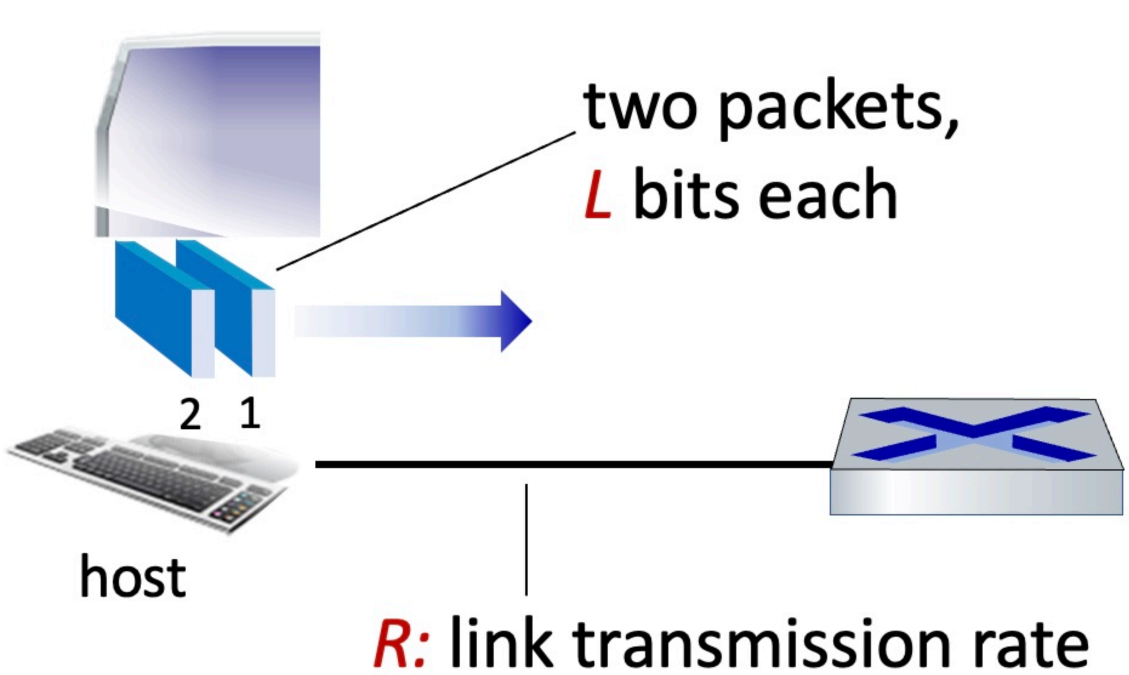
↳ The host breaks the data into smaller chunks, known as packets, of length L.

In addition to data, it'll add some additional information to each chunk of data called packet header.

↳ The host then transmits packets into the access network at transmission rate R
→ wired: send gb/s
→ wireless send Mb/s

↳ Referred as link transmission Rate, link capacity or link bandwidth

$$\text{packet transmission delay} = \text{time needed to transmit } L\text{-bit packet into link} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$



Links: Physical Media

↳ We want to send digital bits over some physical media from sender to reciever

Guided Media

↳ Signals propogate in solid media

copper, fibre, coaxial, twisted pair

Unguided Media

↳ Signals propogate freely

Radio

Guided Media:

Twisted Pair

↳ Refer to actual twisted pair of copper wires.

↳ It is used to carry DSL access network into home.

↳ Susceptible to electromagnetic noise

Coaxial cable:

↳ Two concentric copper conductors

↳ It is used to carry cable-based Access network into home

↳ Bi-directional: Uploads & Downloads at the same time

Fibre Optic Cable

↳ Carry light pulses, each pulse a bit

↳ Have low error rate: immune to electromagnetic noise

Unguided Media (Wireless Radio)

↳ Signals carried in various "bands" in electromagnetic spectrum

↳ Transmissions tend to be broadcasts "half-duplex"

↳ Wireless is a harsh environment for transmitting radio signals as they fade over distance, reflected or blocked by objects at certain frequencies

They are subject to noise generated by motors & other devices that emit RF signals

↳ There are many types of wireless links.

- **Wireless LAN** (WiFi)
  - 10-100's Mbps; 10's of meters
- **wide-area** (e.g., 4G/5G cellular)
  - 100's Mbps (4G/5G) over ~10 Km
- **Bluetooth:** cable replacement
  - short distances, limited rates
- **terrestrial microwave**
  - point-to-point; 45 Mbps channels
- **satellite**
  - up to < 100 Mbps (Starlink) downlink
  - 270 msec end-end delay (geostationary)

  (noticeable propagation delay)

Network Core

↳ Consists of set of routers that are interconnected by a set of communication links.

↳ The internet's core operation is packet switching

Hosts break application level messages into packets.

Network forwards packets from one router to next on a path from source to destination.

↳ There are two key functions performed inside the network core.

1. Forwarding / Switching

   ↳ It is a local action.

   ↳ Its above moving arriving packets from router's input link to appropriate router output link.

   ↳ It is controlled by a forwarding table inside each & every router

   ↳ When a packet arrives, a router will look at the packet header for destination address & look for it in the table

2. Routing

   ↳ It's a global action

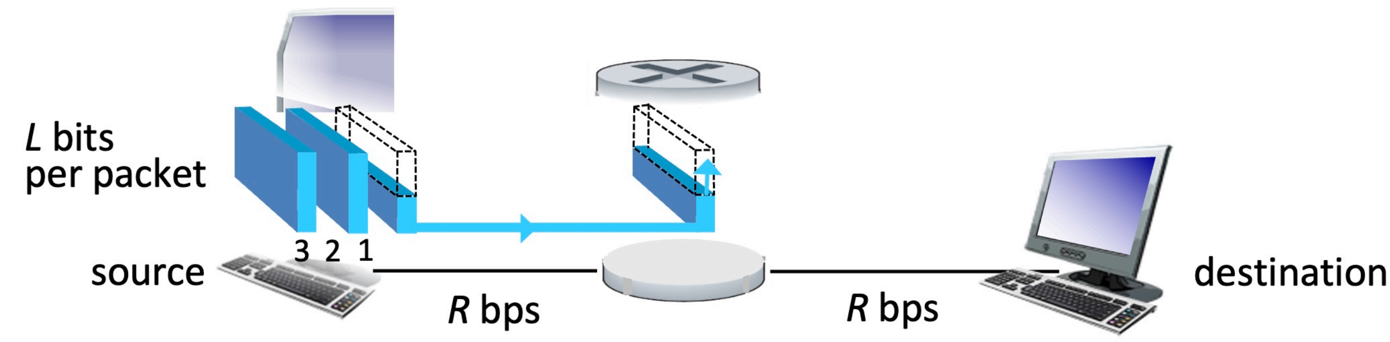
   ↳ It determines source to destination paths taken by packets.

   ↳ Routing algorithm compute these paths & the local per router forwarding tables

      ↳ Once forwarding table is computed, it remains same as long as network topology remains same.

Packet Switching

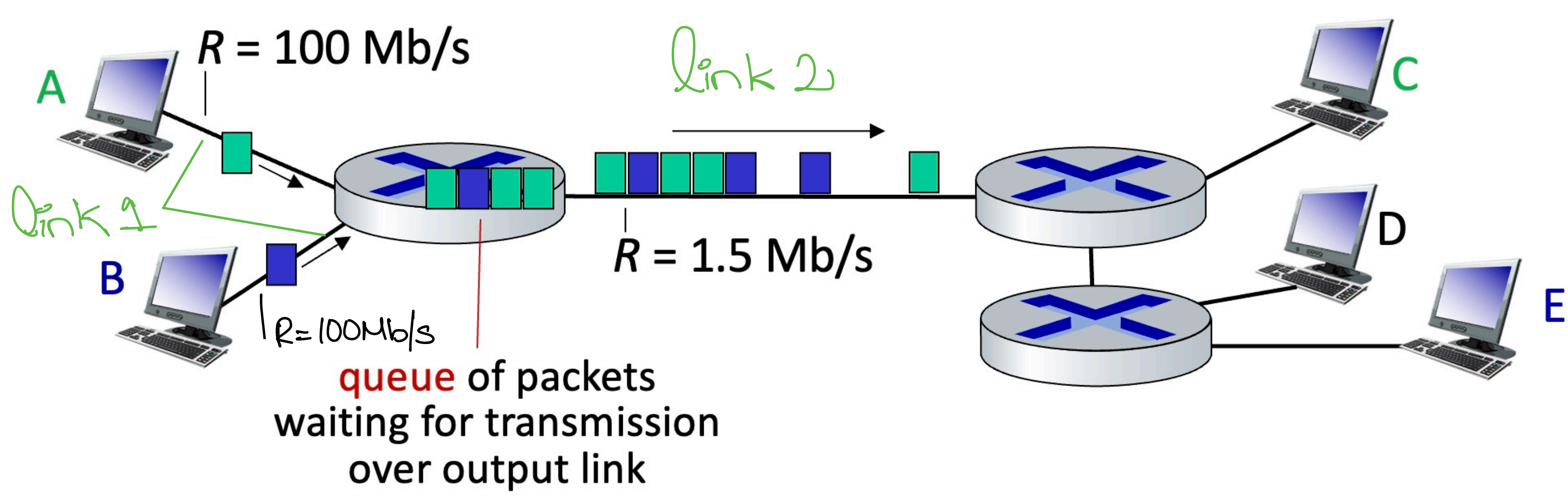
Store & forward



- packet transmission delay: takes L/R seconds to transmit (push out) L-bit packet into link at R bps
- *store and forward: entire* packet must arrive at router before it can be transmitted on next link

One-hop numerical example:
- L = 10 Kbits
- R = 100 Mbps
- one-hop transmission delay = 0.1 msec

Queueing



Assume host A sends packets to host C & host B to host E.

Due to higher R at link 1 than link 2, if A & B are sending alot packets, then a queue of packets is formed at router 1.

Packet queuing and loss: if arrival rate (in bps) to link exceeds transmission rate (bps) of link for some period of time:
- packets will queue, waiting to be transmitted on output link
- packets can be dropped (lost) if memory (buffer) in router fills up

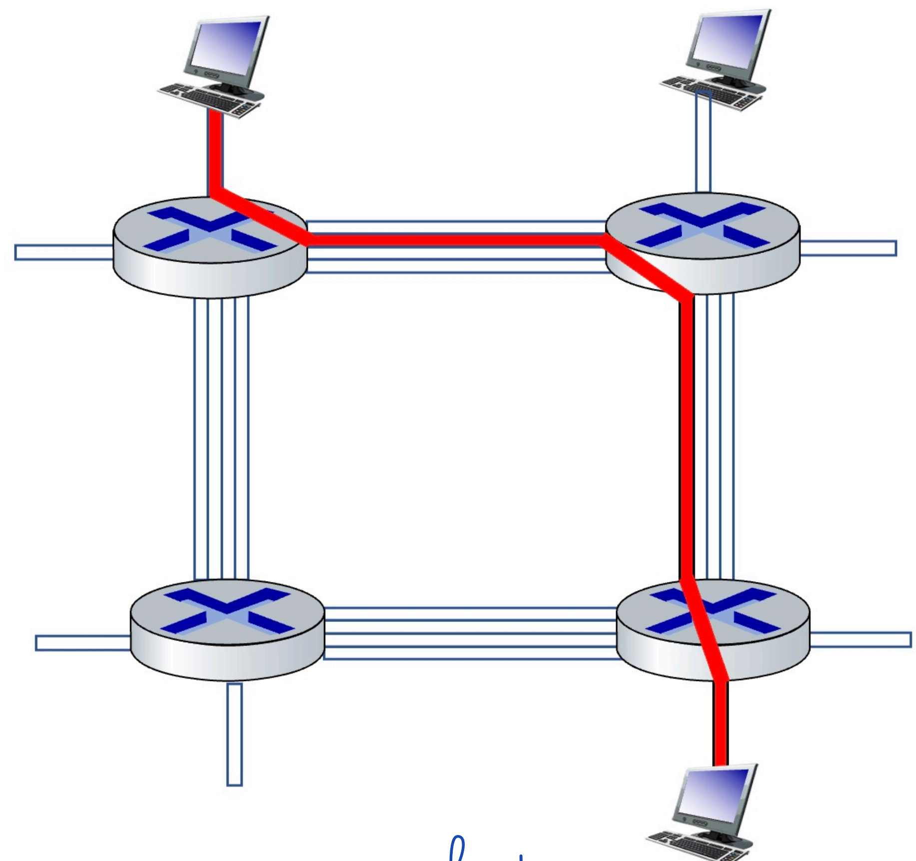
Circuit Switching

↳ Here, call (instead of packets) flow from src to dest

↳ Before a call starts, all of the resources within the network that are going to be needed for that call are allocated to it.

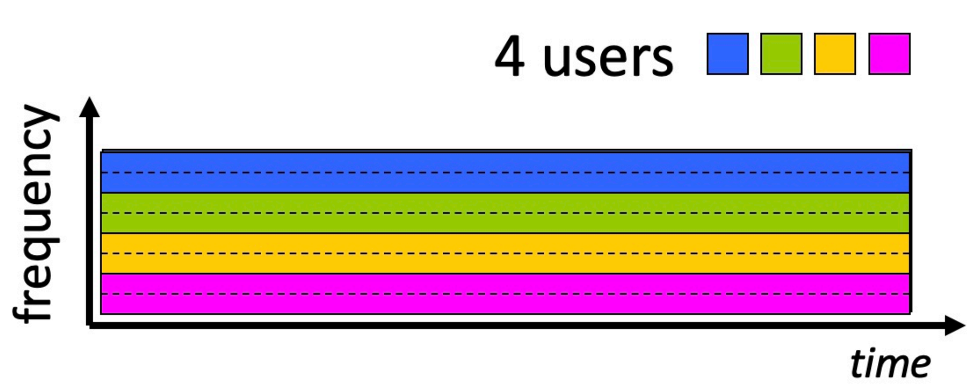
↳ There is only propogation delay & no packet delay or loss

- in diagram, each link has four circuits.
  - call gets 2nd circuit in top link and 1st circuit in right link.
- dedicated resources: no sharing
  - circuit-like (guaranteed) performance
- circuit segment idle if not used by call (no sharing)
- commonly used in traditional telephone networks

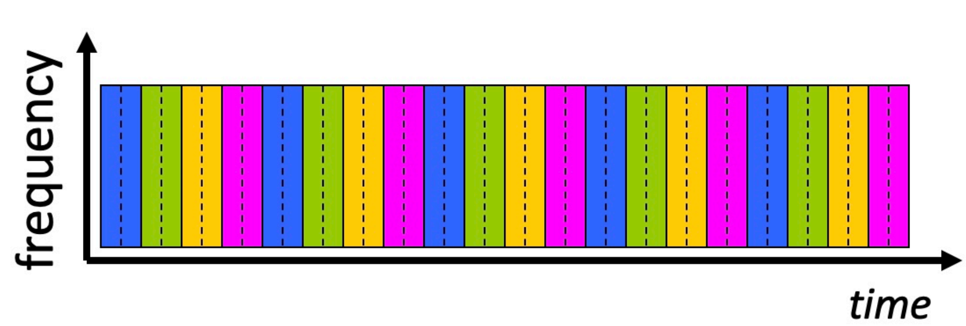


Circuit switching can be done in one of two ways:

Frequency Division Multiplexing (FDM)
- optical, electromagnetic frequencies divided into (narrow) frequency bands
- each call allocated its own band, can transmit at max rate of that narrow band



Time Division Multiplexing (TDM)
- time divided into slots
- each call allocated periodic slot(s), can transmit at maximum rate of (wider) frequency band (only) during its time slot(s)



Is packet switching a “slam dunk winner”?

- great for “bursty” data – sometimes has data to send, but at other times not
  - resource sharing
  - simpler, no call setup
- excessive congestion possible: packet delay and loss due to buffer overflow
  - protocols needed for reliable data transfer, congestion control

• protocols needed for reliable data transfer, congestion control

Performance: loss, delay, throughput